Turning complex decisions into optimal solutions

Management Science

COURSE CODE	CLASS TIME	CLASSROOM	INSTRUCTOR
BD4248	Tue 12:00–13:30	Lecture Hall 207	Kitae Kim

Fall 2026 · School of Business, Korea Aerospace University

WHY MANAGEMENT SCIENCE

How do organizations find the best decision among countless possible options?

Management science uses mathematical modeling and scientific analysis to find optimal solutions to complex decision-making problems, then applies them to real-world management to support rational decisions.

01

The Problem

Limited resources, competing constraints, and too many alternatives to compare by hand

›

02

The Method

Formulate the problem as a mathematical model and solve it with data and computation

›

03

The Decision

An optimal, defensible course of action that can actually be implemented in the business

“If you cannot measure it, you cannot manage it. If you cannot model it, you cannot optimize it.”

COURSE OBJECTIVES

Four things you will take away from this course

01

Why it matters

Understand the necessity of management science and how it is applied in real-world cases.

02

How it works

Understand the various methodologies used in management science, such as optimization.

03

Solving with Python

Acquire the ability to formulate and solve optimization problems using Python.

04

Where it is going

Understand future directions of the field — AI and quantum computing in management science.

Prerequisite · Python — basic programming will make the hands-on work from Week 4 onward far easier to follow.

COURSE INFORMATION

What you need to know before we start

TIME & PLACE

Tuesday 12:00 – 13:30

Lecture Hall 207

CREDITS

3 credits / 3 hours per week

Required major course · Year 2

COURSE CODES

BD4248 — Business Administration

3BD3206 — Business Major

TEXTBOOK & REFERENCES

Modern Management Science, 9th ed.

Se-Heon Kim · Hanbit Academy · January 2023

Lecture slides are built on this textbook.

Buying the book is not required, but it helps a great deal with practice problems and review.

CLASS TYPE

Flipped Learning

Watch → Solve together → Reflect

13 weeks run in this format.

Grading is absolute, not curved.

HOW THIS CLASS RUNS

Every week runs in three stages

Learning starts before you walk into the room. Class time is not for listening to explanations — it is for solving problems together.

BEFORE CLASS — 60 min

Prepare

- Watch the recorded lecture video
- Take the quiz or submit questions

IN CLASS — 75 min

Solve together

- Quiz solutions and core concepts
- Discussion, activities, Python labs

AFTER CLASS — 30 min

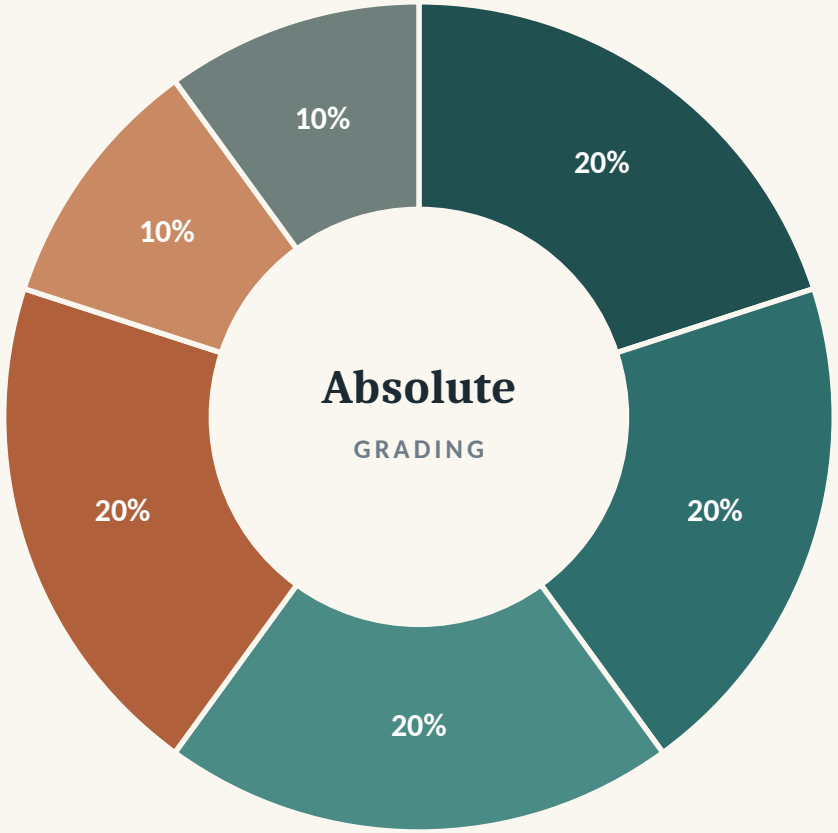
Reflect

- Write the weekly reflective log
- Complete and submit the assignment

If you arrive without having watched the video, those 75 minutes are simply lost. This is the most important commitment in the course.

GRADING

Six components, no curve



Mid-term Exam	20%	Week 8
Final Exam	20%	Week 15
Quiz	20%	Pre-class check · 8 times
Assignment	20%	Weeks 3 · 6 · 10 · 13
Attendance	10%	Checked every session
Participation	10%	Discussion and reflective logs

The participation score includes the reflective logs of students.

GENERATIVE AI POLICY

Generative AI is permitted — conditionally

Permitted

- Brainstorming ideas and drafting outlines
- Sentence correction and grammar checking
- Explaining concepts, generating draft code

Not permitted

- Submitting AI output as your own work
- Using AI without disclosing that you did
- Any use during exams and quizzes

The final work must be produced by you, and any use of AI requires a Generative AI Usage Statement.

The statement records the tool used, the purpose and scope of use, your main prompts, and your own contribution and revisions, followed by a self-check pledge. Its content is not directly reflected in grading — it exists to ensure transparency.

Generative AI reflects social and cultural bias and can produce inaccurate results. If you submit AI-generated errors, the academic responsibility is yours.

YOUR INSTRUCTOR

Questions are always welcome

KK

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Office Hours

No appointment needed — just drop by.

Tuesday	14:00 – 16:00
Wednesday	15:00 – 17:00
Thursday	12:00 – 14:00

SUPPORT FOR STUDENTS WITH DISABILITIES

Under the Rules Pertaining to Disabled Student Support, students are entitled to special conditions or assistance in attending class, receiving materials, completing assignments, and writing exams. Please consult the instructor, the Academic Affairs Team, or the Student Affairs Team before registering for the class.

SEMESTER ROADMAP

Where we are going over 16 weeks

Phase 1

Weeks 1–3

Foundations and linear optimization

Phase 2

Weeks 4–7

Python labs and deeper solution analysis

Phase 3

Weeks 9–11

Integer, network, and multi-criteria models

Phase 4

Weeks 12–14

Game theory and the future of the field

WEEK 8 **Mid-term**

WEEK 15 **Final**

Week 16 is reserved as a make-up session if one is needed.

PHASE 1 · WEEKS 1–3

Building the foundations

Week 1	**Introduction to Management Science**	Course introduction and application cases	Lecture · Questions
Week 2	**Linear optimization and linear programming**	Structure and formulation of an LP	Quiz · Discussion
Week 3	**Development, application, and limitations**	How far LP models can be pushed	Quiz · Assignment #1

From Week 2 onward, every week begins with a pre-class video and a quiz.

PHASE 2 · WEEKS 4–7

Solving in Python, then interpreting the answer

Week 4	**Analysis of LP models using Python**	Hands-on in Google Colab	Lab · Activity
Week 5	**Post-optimality analysis**	Reading what the solution implies	Quiz · Discussion
Week 6	**Sensitivity analysis**	How the optimum responds to change	Quiz · Assignment #2
Week 7	**Analyzing business problems with LP**	Applications across real settings	Quiz · Discussion
Week 8	**Mid-term Exam**	Covers Weeks 1–7	**Exam**

Set up a Google Colab account before the Week 4 lab.

PHASE 3 · WEEKS 9–11

Extending the models beyond linearity

Week 9	**Integer optimization models**	Decisions with integer constraints	Lecture · Questions
Week 10	**Network models and transportation**	Linear and integer models on networks	Lab · Assignment #3
Week 11	**Multi-criteria decision making 1**	Scoring methods and DEA	Quiz · Discussion

From here on, problems have several objectives that genuinely conflict.

PHASE 4 · WEEKS 12–14

Modeling competition, then looking ahead

Week 12	**Multi-criteria decision making 2**	Goal programming	Quiz · Discussion
Week 13	**Game theory**	Analyzing strategic interaction	Quiz · Assignment #4
Week 14	**The future of management science**	Agentic AI and quantum computing	Lecture · Questions
Week 15	**Final Exam**	Focused on Weeks 9–14	Exam

Week 16 is reserved as a make-up session if one is needed.

GROUND RULES

Five commitments for a good semester

1	**Always prepare**	The 60 minutes of video and the quiz before class are where this course starts. Without them, the in-class problem solving does not work.
2	**Ask early**	Submit what you do not understand as a pre-class question, or bring it to office hours — open three days a week.
3	**Bring your laptop**	Python and Google Colab are used from Week 4 onward. You will need a machine you can work on.
4	**Be transparent with AI**	Use it for brainstorming and grammar, but the final work must be yours, and it comes with a usage statement.
5	**Write the reflective log**	The 30-minute log after class counts toward your participation score. Write it the same week.